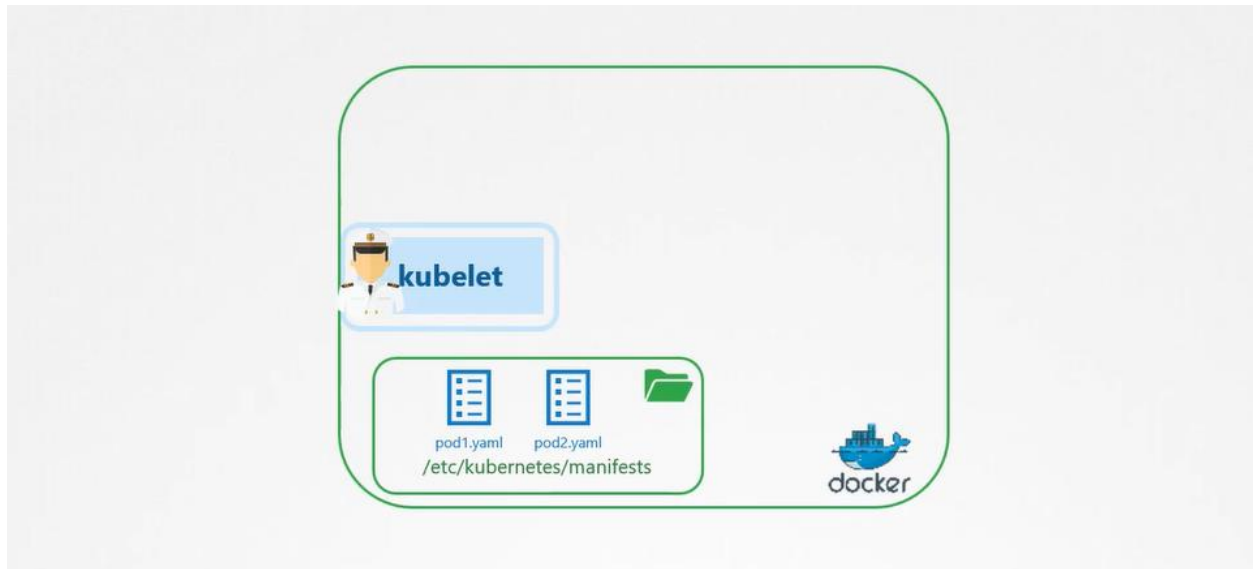


Static Pods



kubelet work at pod level,

How to pass the location? staticPodPath: /etc/kubernetes/manifests

docker ps

kubectl get pod

When kubelet creates a static pod, and it also a part of k8s cluster then it's also creates a mirror object for api-server.

Difference between Static Pod and DaemonSets

Static PODs	DaemonSets
Created by the Kubelet	Created by Kube-API server (DaemonSet Controller)
Deploy Control Plane components as Static Pods	Deploy Monitoring Agents, Logging Agents on nodes
Ignored by the Kube-Scheduler	

Multi Scheduler

```
kubectl get pods -o yaml -n kube-system // kube-scheduler-master
```

By default K8s have their own scheduler (kube-scheduler-master), taking decision in which node, pod should be run.

Why we need another scheduler?

You have a specific application that requires its components deployed on nodes after performing some additional checks. K8s is highly extensible you can write your own k8s scheduler, package it and deploy it as a default scheduler or as an additional scheduler in K8s cluster, all the application can use default scheduler and your specific application can use customized scheduler, can have a multiple scheduler at same time. When creating a Pod can define to which scheduler within spec attribute, e.g. If you want on scheduler your own customization, I want to set my priority e.g. if pod fail default scheduler takes 10 min to reschedule the pod in another node, I want it takes 5 min to schedule.

Deploy additional Scheduler-Kubeadm

/etc/kubernetes/manifests/kube-scheduler.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: kube-scheduler
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-scheduler
    - --address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
    name: kube-scheduler
```

my-custom-scheduler.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: my-custom-scheduler
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-scheduler
    - --address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    - --scheduler-name=my-custom-scheduler
    image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
    name: kube-scheduler
```

step1: copy the kube-scheduler.yaml

step2: change the name of my-custom scheduler

```
kubectl apply -f <my-scheduler.yaml>
```

```
kubectl -n kube-system get pods # check myscheduler is running
```

UC: If multiple schedulers are running on cluster then at a time only one can be active.

```
/etc/kubernetes/manifests/kube-scheduler.yaml
apiVersion: v1
kind: Pod
metadata:
  name: kube-scheduler
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-scheduler
    - --address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
    name: kube-scheduler
```

```
my-custom-scheduler.yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-custom-scheduler
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-scheduler
    - --address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    - --scheduler-name=my-custom-scheduler
    image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
    name: kube-scheduler
```

How to tell our Pod to use our scheduler else it will use default scheduler?

```
kubectl get pods --namespace=kube-system
```

NAME	READY	STATUS	RESTARTS	AGE
coredns-78fcd6894-bk4m1	1/1	Running	0	1h
coredns-78fcd6894-ppr6m	1/1	Running	0	1h
etcd-master	1/1	Running	0	1h
kube-apiserver-master	1/1	Running	0	1h
kube-controller-manager-master	1/1	Running	0	1h
kube-proxy-dgbgv	1/1	Running	0	1h
kube-proxy-fptbr	1/1	Running	0	1h
kube-scheduler-master	1/1	Running	0	1h
my-custom-scheduler	1/1	Running	0	9s
weave-net-4tfpt	2/2	Running	1	1h
weave-net-6j6zs	2/2	Running	1	1h

```
pod-definition.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - image: nginx
    name: nginx
  schedulerName: my-custom-scheduler
```

```
kubectl create -f pod-definition.yaml
```

```

    version: second
  spec:
    serviceAccountName: my-scheduler
    containers:
    - command:
      - /usr/local/bin/kube-scheduler
      - --address=0.0.0.0
      - --leader-elect=false
      - --scheduler-name=my-scheduler
    image: gcr.io/my-gcp-project/my-kube-scheduler:1.0
    livenessProbe:

```

Configuring K8s Scheduler

/etc/kubernetes/manifests/kube-scheduler.yaml

```

apiVersion: v1
kind: Pod
metadata:
  name: kube-scheduler
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-scheduler
    - --address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
    name: kube-scheduler

```

my-custom-scheduler.yaml

```

apiVersion: v1
kind: Pod
metadata:
  name: my-custom-scheduler
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-scheduler
    - --address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    - --scheduler-name=my-custom-scheduler
    - --lock-object-name=my-custom-schedul
    image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
    name: kube-scheduler

```